

Testgatherer: Project Progress Report 1



An interactive web application for managing and displaying user testimonials with scalable architecture and modern technology stack

Introduction

Testigatherer is a comprehensive full-stack web application designed to revolutionize how users create, manage, and share testimonials. The platform enables users to establish personalized testimonial spaces where they can collect feedback, ratings, and experiences from their audience in an organized and visually appealing manner.

Built with modern web technologies, Testigatherer combines the power of React for dynamic frontend experiences with Express.js and Prisma ORM for robust backend operations. The application supports multiple authentication methods including local credentials and Google OAuth, ensuring flexibility and security for all users.

User-Centric Design

Intuitive interface for seamless testimonial management and space customization

Scalable Architecture

Built to handle growing traffic and data with PostgreSQL and cloud infrastructure

Secure Authentication

JWT tokens and OAuth 2.0 integration for protected user sessions

Problem Statement & Proposed Solution

The Challenge

Organizations and individuals struggle with collecting, organizing, and displaying testimonials effectively. Traditional methods lack:

- Centralized management systems for testimonials
- Customizable spaces for different products or services
- Secure authentication and user authorization
- Interactive features for engagement (ratings, likes)
- Scalable infrastructure for growing needs

Existing solutions are often fragmented, requiring multiple tools and platforms to achieve basic testimonial management functionality.

Our Solution

Testigatherer provides a unified platform that addresses these challenges through:

01

Unified Platform

Single application for creating, managing, and displaying testimonials

02

Customizable Spaces

Users create personalized testimonial spaces with unique branding

03

Secure Access

Multi-method authentication with JWT and Google OAuth

04

Interactive Features

Rating systems, likes, and engagement tools built-in

Scope and Objectives

The Testigatherer project encompasses a comprehensive set of objectives designed to deliver a complete testimonial management solution. Our scope includes both immediate deliverables and future extensibility.

- 1

User Authentication & Management

Implement secure registration and login with email/password and Google OAuth integration. Enable profile management and role-based access control.
- 2

Space Creation & Customization

Allow users to create unlimited testimonial spaces with custom names, images, headers, and feedback questions tailored to their needs.
- 3

Testimonial Submission

Enable users to submit rich testimonials including text, images, and ratings. Implement moderation and approval workflows.
- 4

Interactive Features

Develop engagement tools including like/unlike functionality, rating systems, and social sharing capabilities.
- 5

API Development

Build RESTful API endpoints for all operations ensuring scalability and third-party integration potential.

Our primary goal is to create a platform that not only simplifies testimonial management but also provides users with powerful customization and interaction features while maintaining enterprise-grade security and scalability.

Key Features

Multi-Method Authentication

Users can sign up and log in using local credentials or Google OAuth 2.0, providing flexibility and convenience while maintaining security through JWT tokens.

Customizable Spaces

Create personalized testimonial spaces with unique names, logos, headers, and custom questions to gather specific feedback from your audience.

Rich Testimonial Submission

Submit comprehensive testimonials with text content, images, and star ratings, providing detailed feedback and social proof.

Responsive Design

Fully responsive interface built with Tailwind CSS and Framer Motion, ensuring seamless experience across desktop, tablet, and mobile devices.

RESTful API

Well-documented API endpoints enabling easy integration with external systems and third-party platforms for extended functionality.

Technology Stack

Testigatherer leverages a modern, industry-standard technology stack designed for performance, scalability, and developer productivity. Our architecture combines cutting-edge frontend frameworks with robust backend solutions.

Frontend Technologies



React

Component-based UI library for building dynamic single-page applications with virtual DOM for optimal performance



Tailwind CSS

Utility-first CSS framework enabling rapid UI development with responsive, customizable designs



Framer Motion

Animation library for smooth, interactive transitions and engaging user experiences



React Router

Client-side routing for seamless navigation without page reloads

Backend Technologies



Node.js & Express

Lightweight runtime and web framework for efficient server-side JavaScript execution



PostgreSQL

Reliable relational database with ACID compliance for data integrity



Prisma ORM

Type-safe database client with automated query generation and migrations



JWT & OAuth 2.0

Secure authentication with JSON Web Tokens and Google OAuth integration

Development Tools

- Zod for schema validation
- CORS for cross-origin requests
- Dotenv for environment management

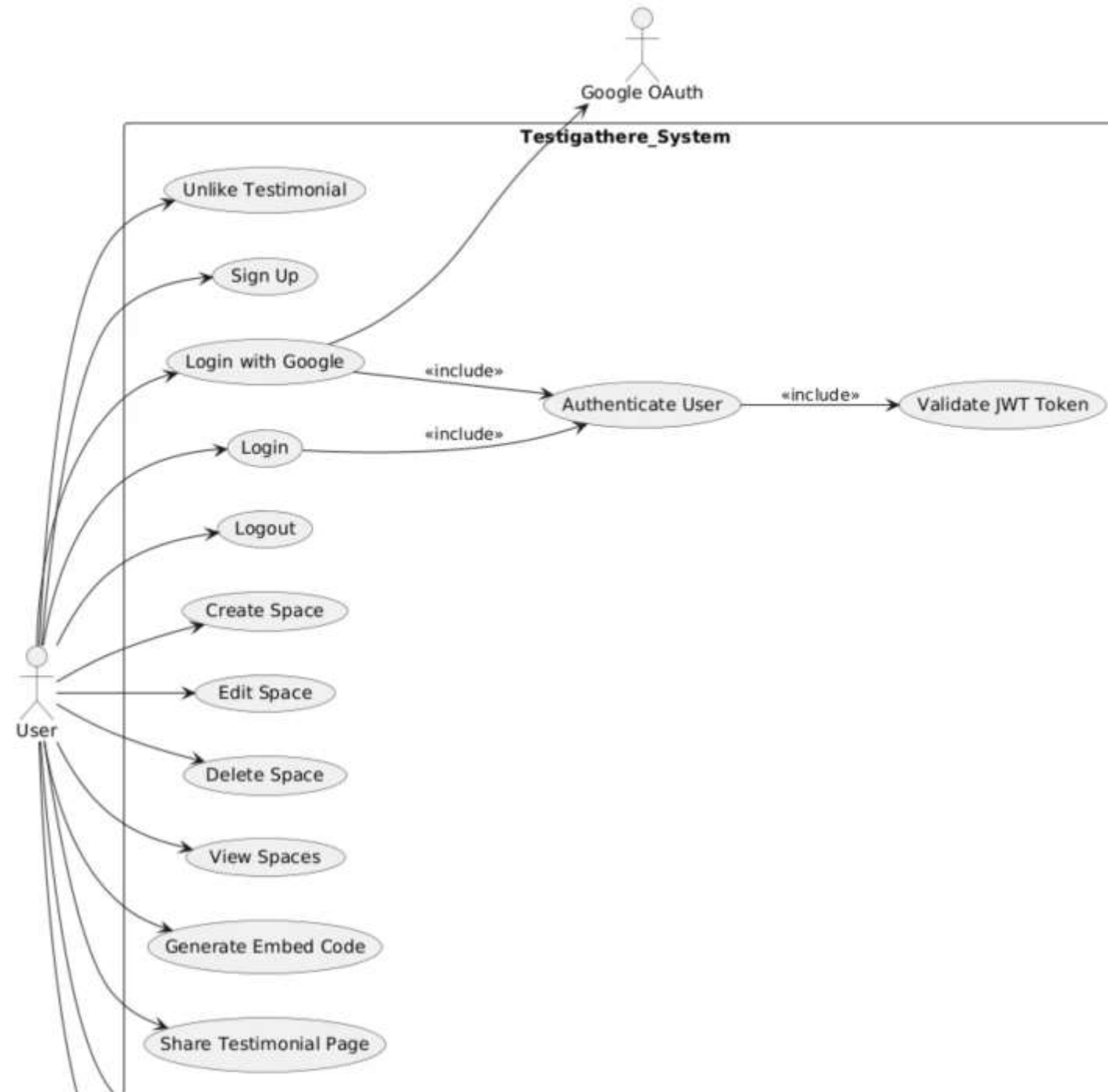
Version Control

- Git for source control
- GitHub for collaboration

Analysis Models

The analysis phase of Testigatherer involved comprehensive modeling to understand system requirements, user interactions, and data flows. These models provide the foundation for our design and implementation.

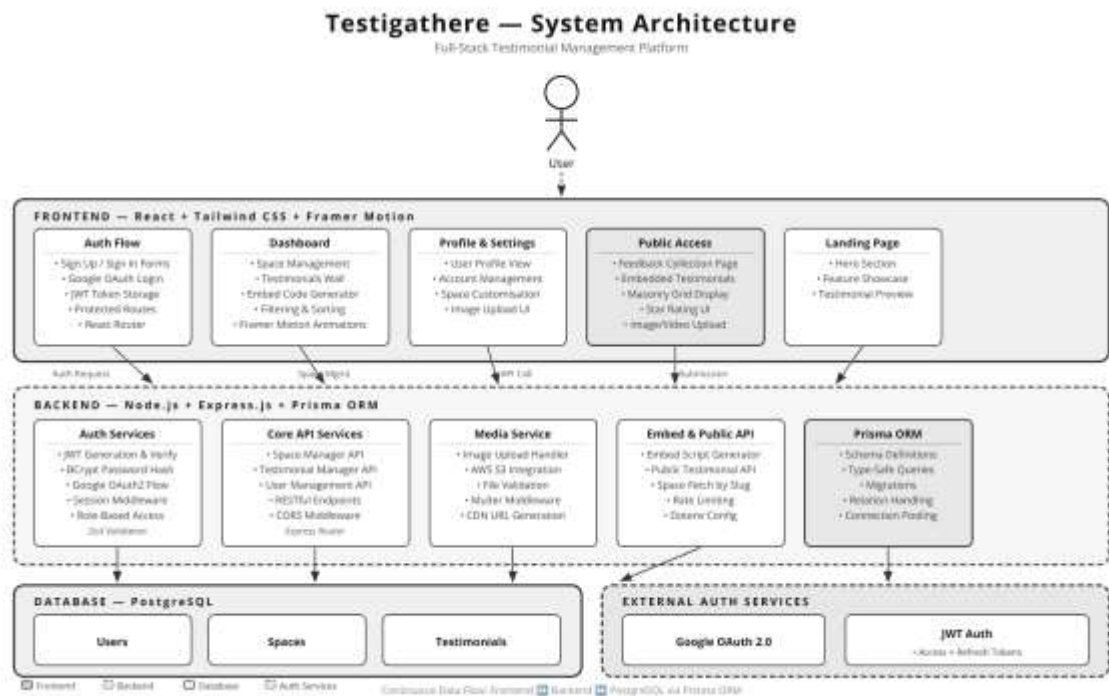
Use Case Diagram



Design Models

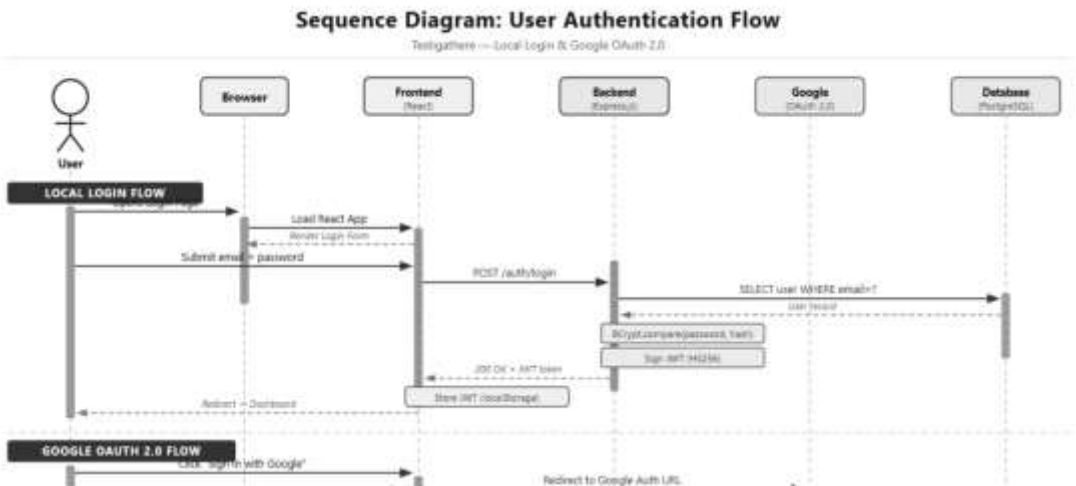
The design models translate our analysis into concrete architectural decisions, defining how components interact and data flows through the system.

System Architecture



Our three-tier architecture separates concerns between presentation, business logic, and data management, ensuring modularity and scalability. The presentation layer handles user interactions, the application layer processes business logic and authentication, and the data layer manages persistent storage.

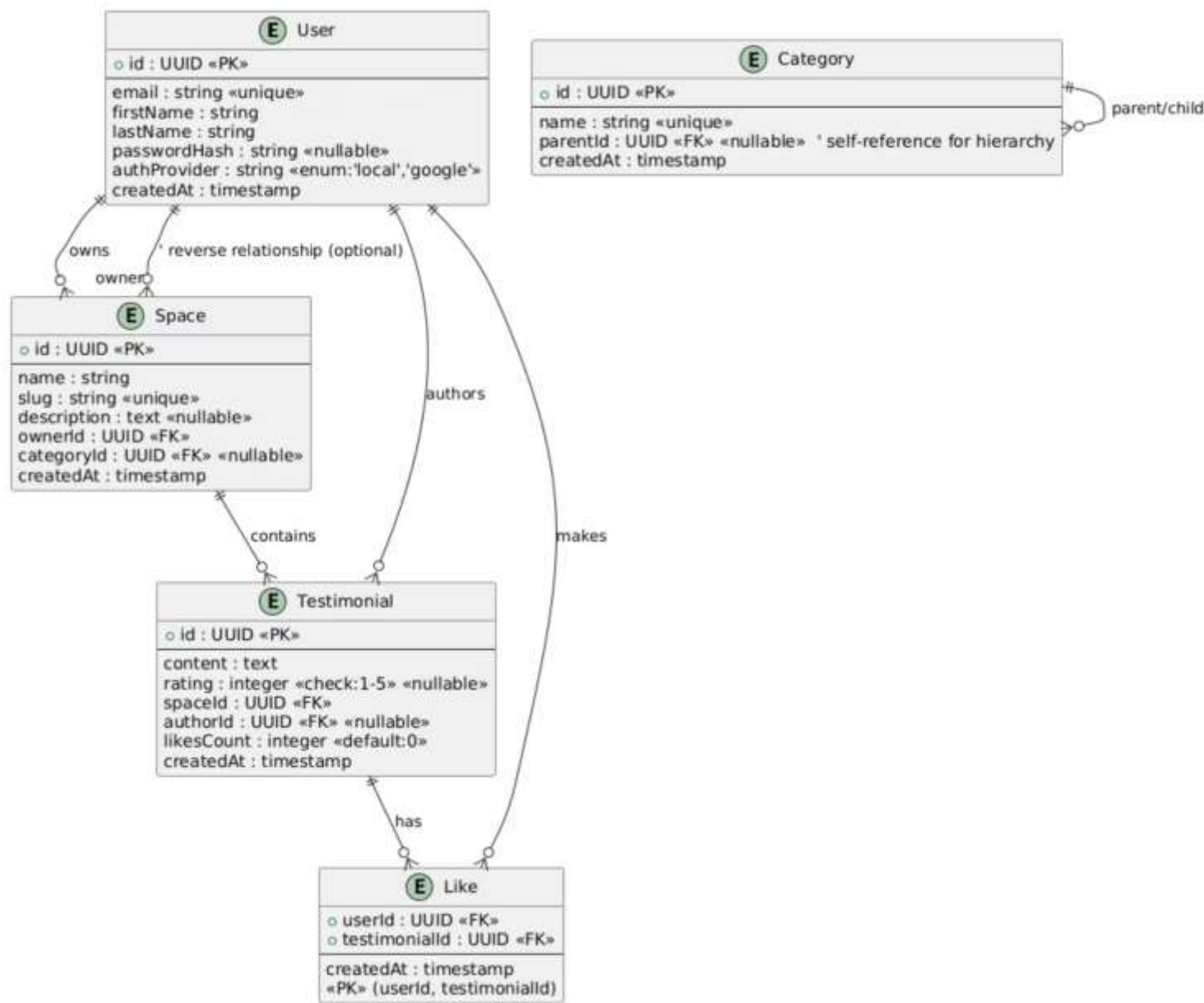
Sequence Diagram: User Authentication Flow



Data Models

The data models define the structure and relationships of information stored in the Testigatherer database, implemented using Prisma ORM with PostgreSQL.

Entity-Relationship Diagram



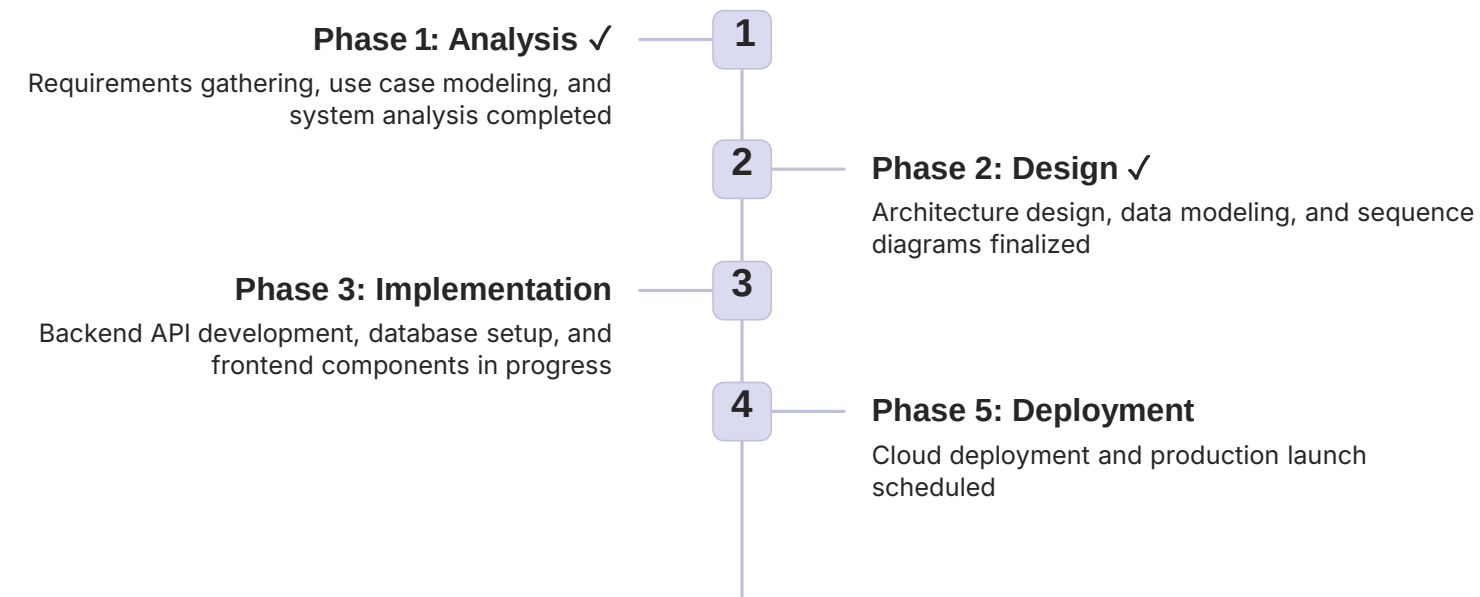
The ER diagram illustrates the core entities and their relationships. Users can create multiple spaces and submit multiple testimonials. Each space can contain multiple testimonials and custom questions, establishing clear ownership and organizational hierarchies.

User Entity

Testimonial Entity

Project Status & Next Steps

This progress report documents the foundational work completed for Testigatherer, including comprehensive analysis, design, and data modeling. The project has established a solid architectural foundation ready for implementation.



Completed Deliverables

- Comprehensive problem statement and solution analysis
- Detailed scope and objectives documentation
- Complete technology stack selection and justification
- Use case diagrams covering all user interactions
- Entity-Relationship diagram with normalized schema
- Sequence diagrams for critical workflows
- System architecture diagram with component interactions
- Database schema design with Prisma ORM integration

Upcoming Milestones

- 01
-
- Backend Development**
API implementation and database integration
- 02
-
- Frontend Development**
React components and UI implementation
- 03
-
- Production Deployment**
Cloud hosting and performance optimization

The Testigatherer project is on track to deliver a comprehensive testimonial management platform that combines security, scalability, and user-friendly design. Our thorough analysis and design phase ensures a solid foundation for successful implementation and deployment.